Online Retail II Multi-Year Sales and Returns Analytics

SQL Phase 2 Capstone Project - Prepared for Shraddha Yadav

1. Domain, Dataset and End Consumer

Domain: Employment & Poverty / Consumer Access - multi-year retail transactions and returns

Dataset: online_retail_II.csv

Cleaned SQLite table name: online_retail_ii

Rows used in project database: 5,000. Columns: 15.

End consumer: A retail operations and customer-access analyst who needs to understand affordability, product performance, customer reach, return risk, and operational planning issues from transaction/product data.

Important approval note: this pack is approval-first. Send the dataset approval email before treating the SQL/report as final submission.

2. Five Analytical Perspectives

1. Revenue and Market Coverage
2. Customer and Access Analysis
3. Product Demand and Catalogue Analysis
4. Returns, Cancellations and Data Quality
5. Time Trends and Operational Planning

3. SQL Concept Coverage

Concept	Number of queries
SELECT	7
WHERE	18
GROUP BY	25
CTE	9
VIEW	5
Window Function	11

4. Query Outputs and Interpretations

Q01. What are the first 10 transaction records?

Perspective: Revenue and Market Coverage | Difficulty: Easy | SQL concept: SELECT

SQL used:

```
SELECT invoice_no, stock_code, description, quantity, unit_price, country FROM online_retail_ii LIMIT 10;
```

Output sample:

invoice_no	stock_code	description	quantity	unit_price	country
489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	6.95	United Kingdom
489460	22273	FELTCRAFT DOLL	6	2.95	United Kingdom

		MOLLY			
489519	84884A	ANT WHITE WIRE HEART SPIRAL	5	3.95	United Kingdom
489529	20673	STRAWBERRIES PRINT BOWL	4	1.25	United Kingdom
489536	21934	SKULL SHOULDER BAG	2	1.65	United Kingdom

Interpretation: Shows a quick sample of the transaction data.

Q02. Which transactions are from the United Kingdom?

Perspective: Revenue and Market Coverage | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT invoice_no, description, quantity, unit_price, country FROM online_retail_ii WHERE country = 'United Kingdom' LIMIT 20;
```

Output sample:

invoice_no	description	quantity	unit_price	country
489434	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	6.95	United Kingdom
489460	FELTCRAFT DOLL MOLLY	6	2.95	United Kingdom
489519	ANT WHITE WIRE HEART SPIRAL	5	3.95	United Kingdom
489529	STRAWBERRIES PRINT BOWL	4	1.25	United Kingdom
489536	SKULL SHOULDER BAG	2	1.65	United Kingdom

Interpretation: Filters the dataset to UK transactions.

Q03. Which lines have a positive revenue value?

Perspective: Revenue and Market Coverage | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT invoice_no, description, line_revenue FROM online_retail_ii WHERE line_revenue > 0 ORDER BY line_revenue DESC LIMIT 20;
```

Output sample:

invoice_no	description	line_revenue
531411	AMAZON FEE	6706.71
574294	PAPER CHAIN KIT 50'S CHRISTMAS	2601.0
539101	GROW A FLYTRAP OR SUNFLOWER IN TIN	2256.0
518735	SET OF 3 RED STRIPE BASKETS	1570.94
532140	JUMBO BAG RED RETROSPOT	1320.0

Interpretation: Finds high-value paid transaction lines.

Q04. Which countries appear in the dataset?

Perspective: Revenue and Market Coverage | Difficulty: Easy | SQL concept: SELECT

SQL used:

```
SELECT DISTINCT country FROM online_retail_ii WHERE country <> '' ORDER BY country LIMIT 50;
```

Output sample:

country
Australia
Austria
Belgium
Brazil
Canada

Interpretation: Lists different customer countries.

Q05. Which transactions have unit prices above 20?

Perspective: Revenue and Market Coverage | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT invoice_no, description, unit_price, country FROM online_retail_ii WHERE unit_price > 20 ORDER BY unit_price DESC LIMIT 20;
```

Output sample:

invoice_no	description	unit_price	country
531411	AMAZON FEE	6706.71	United Kingdom
498133	DOTCOM POSTAGE	543.57	United Kingdom
534859	DOTCOM POSTAGE	518.79	United Kingdom
C518904	Bank Charges	402.69	United Kingdom
C525910	Manual	311.99	United Kingdom

Interpretation: Shows higher-priced items that may drive revenue.

Q06. Which countries generate the highest total revenue?

Perspective: Revenue and Market Coverage | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT country, ROUND(SUM(line_revenue),2) AS total_revenue, COUNT(*) AS transaction_lines FROM online_retail_ii GROUP BY country ORDER BY total_revenue DESC LIMIT 15;
```

Output sample:

country	total_revenue	transaction_lines
United Kingdom	83402.03	4603
EIRE	2668.39	81
Germany	2563.35	84
Netherlands	2252.08	23
France	1677.72	72

Interpretation: Groups by country to compare revenue contribution.

Q07. What is the average order line value by country?

Perspective: Revenue and Market Coverage | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT country, ROUND(AVG(line_revenue),2) AS avg_line_revenue, COUNT(*) AS lines FROM online_retail_ii WHERE line_revenue > 0 GROUP BY country HAVING COUNT(*) >= 5 ORDER BY avg_line_revenue DESC LIMIT 15;
```

Output sample:

country	avg_line_revenue	lines
Netherlands	97.92	23
Australia	89.7	9
Austria	70.42	5
Sweden	55.91	7
Spain	38.31	14

Interpretation: Compares typical transaction values across countries.

Q08. Which countries have the most unique customers?

Perspective: Revenue and Market Coverage | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT country, COUNT(DISTINCT customer_id) AS unique_customers FROM online_retail_ii WHERE customer_id IS NOT NULL GROUP BY country ORDER BY unique_customers DESC LIMIT 15;
```

Output sample:

country	unique_customers
United Kingdom	1829
Germany	46
France	33
Switzerland	12
Spain	10

Interpretation: Shows customer reach by country.

Q09. Which countries have the highest total quantity sold?

Perspective: Revenue and Market Coverage | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT country, SUM(quantity) AS total_quantity FROM online_retail_ii WHERE quantity > 0 GROUP BY country ORDER BY total_quantity DESC LIMIT 15;
```

Output sample:

country	total_quantity
United Kingdom	48040
Netherlands	1458
France	1363
EIRE	1307
Germany	1075

Interpretation: Identifies markets with high sales volume.

Q10. Which countries have unusually low customer coverage?

Perspective: Revenue and Market Coverage | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT country, COUNT(DISTINCT customer_id) AS unique_customers, COUNT(*) AS lines FROM online_retail_ii GROUP BY country HAVING unique_customers < 10 ORDER BY lines DESC LIMIT 20;
```

Output sample:

country	unique_customers	lines
EIRE	3	81
Netherlands	6	23
Belgium	7	12
Australia	5	10
Italy	8	8

Interpretation: Flags countries with fewer identified customers.

Q11. Which top countries contribute most to overall revenue share?

Perspective: Revenue and Market Coverage | Difficulty: Difficult | SQL concept: CTE

SQL used:

```
WITH country_revenue AS (SELECT country, SUM(line_revenue) AS revenue FROM online_retail_ii WHERE line_revenue > 0 GROUP BY country), total_revenue AS (SELECT SUM(revenue) AS total_rev FROM country_revenue) SELECT country, ROUND(revenue,2) AS revenue, ROUND(100.0*revenue/total_rev,2) AS pct_of_total FROM country_revenue, total_revenue ORDER BY revenue DESC LIMIT 15;
```

Output sample:

country	revenue	pct_of_total
United Kingdom	91132.63	86.7

EIRE	2712.64	2.58
Germany	2570.13	2.45
Netherlands	2252.08	2.14
France	1766.67	1.68

Interpretation: Uses CTEs to calculate country revenue and percentage share.

Q12. How do countries rank by revenue?

Perspective: Revenue and Market Coverage | Difficulty: Difficult | SQL concept: Window Function

SQL used:

```
SELECT country, ROUND(SUM(line_revenue),2) AS total_revenue, RANK() OVER (ORDER BY SUM(line_revenue) DESC) AS revenue_rank
FROM online_retail_ii WHERE line_revenue > 0 GROUP BY country ORDER BY revenue_rank LIMIT 15;
```

Output sample:

country	total_revenue	revenue_rank
United Kingdom	91132.63	1
EIRE	2712.64	2
Germany	2570.13	3
Netherlands	2252.08	4
France	1766.67	5

Interpretation: Ranks countries by revenue using a window function.

Q13. What is each country revenue compared with the previous rank?

Perspective: Revenue and Market Coverage | Difficulty: Difficult | SQL concept: Window Function

SQL used:

```
WITH ranked AS (SELECT country, SUM(line_revenue) AS revenue, RANK() OVER (ORDER BY SUM(line_revenue) DESC) AS rnk FROM
online_retail_ii WHERE line_revenue > 0 GROUP BY country) SELECT country, ROUND(revenue,2) AS revenue, rnk, ROUND(revenue -
LAG(revenue) OVER (ORDER BY rnk),2) AS gap_from_previous FROM ranked ORDER BY rnk LIMIT 15;
```

Output sample:

country	revenue	rnk	gap_from_previous
United Kingdom	91132.63	1	None
EIRE	2712.64	2	-88419.99
Germany	2570.13	3	-142.51
Netherlands	2252.08	4	-318.05
France	1766.67	5	-485.41

Interpretation: Uses LAG to compare country revenue gaps.

Q14. Create a reusable country revenue view and show the top countries.

Perspective: Revenue and Market Coverage | Difficulty: Difficult | SQL concept: VIEW

SQL used:

```
SELECT country, ROUND(total_revenue,2) AS total_revenue, transaction_lines FROM v_country_revenue ORDER BY total_revenue DESC
LIMIT 15;
```

Output sample:

country	total_revenue	transaction_lines
United Kingdom	91132.63	4496
EIRE	2712.64	79
Germany	2570.13	83
Netherlands	2252.08	23
France	1766.67	70

Interpretation: Uses a reusable view for country revenue analysis.

Q15. Which countries are high revenue but have fewer customers?

Perspective: Revenue and Market Coverage | Difficulty: Difficult | SQL concept: CTE

SQL used:

```
WITH country_metrics AS (SELECT country, SUM(line_revenue) AS revenue, COUNT(DISTINCT customer_id) AS customers FROM
online_retail_ii WHERE line_revenue > 0 GROUP BY country), ranked AS (SELECT *, RANK() OVER (ORDER BY revenue DESC) AS
revenue_rank FROM country_metrics) SELECT country, ROUND(revenue,2) AS revenue, customers, revenue_rank FROM ranked WHERE
revenue_rank <= 20 AND customers < 50 ORDER BY revenue_rank;
```

Output sample:

country	revenue	customers	revenue_rank
EIRE	2712.64	3	2
Germany	2570.13	46	3
Netherlands	2252.08	6	4
France	1766.67	33	5
Australia	807.28	5	6

Interpretation: Combines revenue rank with customer count for priority review.

Q16. Which rows have a known customer ID?

Perspective: Customer and Access Analysis | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT invoice_no, customer_id, country, line_revenue FROM online_retail_ii WHERE customer_id IS NOT NULL LIMIT 20;
```

Output sample:

invoice_no	customer_id	country	line_revenue
489434	13085.0	United Kingdom	83.4
489460	16167.0	United Kingdom	17.700000000000003
489519	17700.0	United Kingdom	19.75
489529	17984.0	United Kingdom	5.0
489536	16393.0	United Kingdom	3.3

Interpretation: Checks records linked to identified customers.

Q17. Which rows have missing customer ID?

Perspective: Customer and Access Analysis | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT invoice_no, country, description, line_revenue FROM online_retail_ii WHERE customer_id IS NULL LIMIT 20;
```

Output sample:

invoice_no	country	description	line_revenue
489548	United Kingdom	PAPER CHAIN KIT 50'S CHRISTMAS	8.850000000000001
489597	United Kingdom	SANDWICH BATH SPONGE	7.709999999999999
489597	United Kingdom	CHOCOLATE HOT WATER BOTTLE	11.26
489597	United Kingdom	CHARLIE & LOLA WASTEPAPER BIN BLUE	7.78
489601	United Kingdom	BROWN PIRATE TREASURE CHEST	3.43

Interpretation: Shows transactions without a customer identifier.

Q18. Which customers bought from countries outside the UK?

Perspective: Customer and Access Analysis | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT invoice_no, customer_id, country, description FROM online_retail_ii WHERE country <> 'United Kingdom' AND customer_id
IS NOT NULL LIMIT 20;
```

Output sample:

invoice_no	customer_id	country	description
489883	12437.0	France	RED DAISY PAPER LAMPSHADE
490458	12583.0	France	BLUE SPOTTY CUP
490751	12358.0	Austria	EDWARDIAN PARASOL NATURAL
490964	14646.0	Netherlands	SET/2 RED SPOTTY TEA TOWELS
491340	12979.0	United Arab Emirates	UNION JACK HOT WATER BOTTLE

Interpretation: Filters non-UK customer transactions.

Q19. Which customer IDs appear in the data?

Perspective: Customer and Access Analysis | Difficulty: Easy | SQL concept: SELECT

SQL used:

```
SELECT DISTINCT customer_id FROM online_retail_ii WHERE customer_id IS NOT NULL ORDER BY customer_id LIMIT 50;
```

Output sample:

customer_id
12347.0
12349.0
12351.0
12356.0
12357.0

Interpretation: Lists customer IDs for exploration.

Q20. Which customers have large individual line purchases?

Perspective: Customer and Access Analysis | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT invoice_no, customer_id, country, line_revenue FROM online_retail_ii WHERE line_revenue > 100 ORDER BY line_revenue
DESC LIMIT 20;
```

Output sample:

invoice_no	customer_id	country	line_revenue
531411	None	United Kingdom	6706.71
574294	16333.0	United Kingdom	2601.0
539101	16029.0	United Kingdom	2256.0
518735	18102.0	United Kingdom	1570.94
532140	15838.0	United Kingdom	1320.0

Interpretation: Identifies higher-value customer transaction lines.

Q21. Which customers have the highest total spend?

Perspective: Customer and Access Analysis | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT customer_id, ROUND(SUM(line_revenue),2) AS total_spend, COUNT(DISTINCT invoice_no) AS invoices FROM online_retail_ii
WHERE customer_id IS NOT NULL AND line_revenue > 0 GROUP BY customer_id ORDER BY total_spend DESC LIMIT 20;
```

Output sample:

customer_id	total_spend	invoices
16029.0	2805.5	5
18102.0	2611.86	5
16333.0	2601.0	1

17450.0	2470.4	4
14646.0	2141.73	17

Interpretation: Aggregates customer spend.

Q22. Which customers bought the most quantity?

Perspective: Customer and Access Analysis | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT customer_id, SUM(quantity) AS total_quantity, COUNT(*) AS lines FROM online_retail_ii WHERE customer_id IS NOT NULL AND quantity > 0 GROUP BY customer_id ORDER BY total_quantity DESC LIMIT 20;
```

Output sample:

customer_id	total_quantity	lines
16029.0	2750	5
14646.0	1393	17
13694.0	1128	4
16333.0	1020	1
17450.0	914	4

Interpretation: Finds high-volume customers.

Q23. Which countries have the highest missing customer ID rate?

Perspective: Customer and Access Analysis | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT country, COUNT(*) AS total_lines, SUM(CASE WHEN customer_id IS NULL THEN 1 ELSE 0 END) AS missing_customer_rows, ROUND(100.0*SUM(CASE WHEN customer_id IS NULL THEN 1 ELSE 0 END)/COUNT(*),2) AS missing_pct FROM online_retail_ii GROUP BY country HAVING total_lines >= 20 ORDER BY missing_pct DESC LIMIT 20;
```

Output sample:

country	total_lines	missing_customer_rows	missing_pct
United Kingdom	4603	1121	24.35
EIRE	81	12	14.81
France	72	2	2.78
Netherlands	23	0	0.0
Germany	84	0	0.0

Interpretation: Measures customer data completeness by country.

Q24. What is average customer spend by country?

Perspective: Customer and Access Analysis | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT country, ROUND(SUM(line_revenue)/NULLIF(COUNT(DISTINCT customer_id),0),2) AS revenue_per_customer, COUNT(DISTINCT customer_id) AS customers FROM online_retail_ii WHERE customer_id IS NOT NULL AND line_revenue > 0 GROUP BY country ORDER BY revenue_per_customer DESC LIMIT 20;
```

Output sample:

country	revenue_per_customer	customers
EIRE	835.3	3
Netherlands	375.35	6
Australia	161.46	5
Singapore	156.6	1
Lithuania	127.4	1

Interpretation: Compares revenue per identified customer.

Q25. Which customers have many invoices?

Perspective: Customer and Access Analysis | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT customer_id, COUNT(DISTINCT invoice_no) AS invoice_count, ROUND(SUM(line_revenue),2) AS total_spend FROM
online_retail_ii WHERE customer_id IS NOT NULL GROUP BY customer_id HAVING invoice_count >= 2 ORDER BY invoice_count DESC,
total_spend DESC LIMIT 20;
```

Output sample:

customer_id	invoice_count	total_spend
17841.0	59	350.62
14911.0	49	1034.19
14606.0	38	192.97
12748.0	35	194.19
15311.0	32	396.97

Interpretation: Finds repeat customers.

Q26. Rank customers by total spend.

Perspective: Customer and Access Analysis | Difficulty: Difficult | SQL concept: Window Function

SQL used:

```
WITH customer_spend AS (SELECT customer_id, SUM(line_revenue) AS total_spend FROM online_retail_ii WHERE customer_id IS NOT
NULL AND line_revenue > 0 GROUP BY customer_id) SELECT customer_id, ROUND(total_spend,2) AS total_spend, RANK() OVER (ORDER BY
total_spend DESC) AS spend_rank FROM customer_spend ORDER BY spend_rank LIMIT 20;
```

Output sample:

customer_id	total_spend	spend_rank
16029.0	2805.5	1
18102.0	2611.86	2
16333.0	2601.0	3
17450.0	2470.4	4
14646.0	2141.73	5

Interpretation: Ranks customers by spend.

Q27. Which customers are above the average customer spend?

Perspective: Customer and Access Analysis | Difficulty: Difficult | SQL concept: CTE

SQL used:

```
WITH customer_spend AS (SELECT customer_id, SUM(line_revenue) AS total_spend FROM online_retail_ii WHERE customer_id IS NOT
NULL AND line_revenue > 0 GROUP BY customer_id), avg_spend AS (SELECT AVG(total_spend) AS avg_total_spend FROM customer_spend)
SELECT customer_id, ROUND(total_spend,2) AS total_spend, ROUND(avg_total_spend,2) AS average_spend FROM customer_spend,
avg_spend WHERE total_spend > avg_total_spend ORDER BY total_spend DESC LIMIT 20;
```

Output sample:

customer_id	total_spend	average_spend
16029.0	2805.5	43.38
18102.0	2611.86	43.38
16333.0	2601.0	43.38
17450.0	2470.4	43.38
14646.0	2141.73	43.38

Interpretation: Uses CTEs to compare customers against average.

Q28. Rank customers within each country by spend.

Perspective: Customer and Access Analysis | Difficulty: Difficult | SQL concept: Window Function

SQL used:

```
WITH spend AS (SELECT country, customer_id, SUM(line_revenue) AS total_spend FROM online_retail_ii WHERE customer_id IS NOT NULL AND line_revenue > 0 GROUP BY country, customer_id) SELECT country, customer_id, ROUND(total_spend,2) AS total_spend, ROW_NUMBER() OVER (PARTITION BY country ORDER BY total_spend DESC) AS country_customer_rank FROM spend ORDER BY country, country_customer_rank LIMIT 50;
```

Output sample:

country	customer_id	total_spend	country_customer_rank
Australia	12415.0	648.78	1
Australia	12388.0	71.4	2
Australia	12389.0	44.55	3
Australia	12393.0	27.8	4
Australia	12422.0	14.75	5

Interpretation: Uses partitioned ranking to find leading customers in each country.

Q29. Use the customer value view to identify top customers.

Perspective: Customer and Access Analysis | Difficulty: Difficult | SQL concept: VIEW

SQL used:

```
SELECT customer_id, ROUND(total_spend,2) AS total_spend, invoice_count, country_count FROM v_customer_value ORDER BY total_spend DESC LIMIT 20;
```

Output sample:

customer_id	total_spend	invoice_count	country_count
16029.0	2805.5	5	1
18102.0	2611.86	5	1
16333.0	2601.0	1	1
17450.0	2470.4	4	1
14646.0	2141.73	17	1

Interpretation: Uses a reusable customer value view.

Q30. Which countries have low customer coverage but high spend?

Perspective: Customer and Access Analysis | Difficulty: Difficult | SQL concept: CTE

SQL used:

```
WITH country_metrics AS (SELECT country, COUNT(DISTINCT customer_id) AS customers, SUM(line_revenue) AS revenue FROM online_retail_ii WHERE line_revenue > 0 GROUP BY country) SELECT country, customers, ROUND(revenue,2) AS revenue FROM country_metrics WHERE customers < 20 AND revenue > (SELECT AVG(revenue) FROM country_metrics) ORDER BY revenue DESC;
```

Output sample:

country	customers	revenue

Interpretation: Highlights countries with concentrated sales but few customers.

Q31. What are sample products in the data?

Perspective: Product Demand and Catalogue Analysis | Difficulty: Easy | SQL concept: SELECT

SQL used:

```
SELECT stock_code, description, quantity, unit_price FROM online_retail_ii LIMIT 20;
```

Output sample:

stock_code	description	quantity	unit_price
85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	6.95
22273	FELTCRAFT DOLL MOLLY	6	2.95
84884A	ANT WHITE WIRE HEART SPIRAL	5	3.95
20673	STRAWBERRIES PRINT BOWL	4	1.25
21934	SKULL SHOULDER BAG	2	1.65

Interpretation: Shows product line examples.

Q32. Which products have negative quantities?

Perspective: Product Demand and Catalogue Analysis | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT invoice_no, stock_code, description, quantity FROM online_retail_ii WHERE quantity < 0 LIMIT 20;
```

Output sample:

invoice_no	stock_code	description	quantity
C491012	21231	SWEETHEART CERAMIC TRINKET BOX	-2
C491190	21901	KEY FOB , BACK DOOR	-24
C493191	20871	OPULENT VELVET SET/3 CANDLES	-1
C493869	85044	RED REINDEER STRING OF 20 LIGHTS	-1
C494452	85047	WHITE BEADED GARLAND STRING 20LIGHT	-1

Interpretation: Shows likely returns or corrections.

Q33. Which products have zero or missing unit price?

Perspective: Product Demand and Catalogue Analysis | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT invoice_no, stock_code, description, unit_price FROM online_retail_ii WHERE unit_price IS NULL OR unit_price <= 0 LIMIT 20;
```

Output sample:

invoice_no	stock_code	description	unit_price
495860	16254		0.0
496903	21176		0.0
496955	21772		0.0
498036	16161N		0.0
498693	84653		0.0

Interpretation: Finds free or invalid price records.

Q34. Which products are high priced?

Perspective: Product Demand and Catalogue Analysis | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT stock_code, description, unit_price FROM online_retail_ii WHERE unit_price > 10 ORDER BY unit_price DESC LIMIT 20;
```

Output sample:

stock_code	description	unit_price
AMAZONFEE	AMAZON FEE	6706.71
DOT	DOTCOM POSTAGE	543.57
DOT	DOTCOM POSTAGE	518.79
BANK CHARGES	Bank Charges	402.69
M	Manual	311.99

Interpretation: Filters higher-price products.

Q35. Which unique stock codes exist?

Perspective: Product Demand and Catalogue Analysis | Difficulty: Easy | SQL concept: SELECT

SQL used:

```
SELECT DISTINCT stock_code, description FROM online_retail_ii WHERE stock_code <> '' ORDER BY stock_code LIMIT 50;
```

Output sample:

stock_code	description
10002	INFLATABLE POLITICAL GLOBE
10120	DOGGY RUBBER
10125	MINI FUNKY DESIGN TAPES
10133	COLOURING PENCILS BROWN TUBE
10138	ASSORTED COLOUR JUMBO PEN

Interpretation: Lists product codes.

Q36. Which products generate the highest revenue?

Perspective: Product Demand and Catalogue Analysis | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT stock_code, description, ROUND(SUM(line_revenue),2) AS total_revenue, SUM(quantity) AS quantity_sold FROM
online_retail_ii WHERE line_revenue > 0 GROUP BY stock_code, description ORDER BY total_revenue DESC LIMIT 20;
```

Output sample:

stock_code	description	total_revenue	quantity_sold
AMAZONFEE	AMAZON FEE	6706.71	1
22086	PAPER CHAIN KIT 50'S CHRISTMAS	2988.41	1159
22693	GROW A FLYTRAP OR SUNFLOWER IN TIN	2286.0	2424
22423	REGENCY CAKESTAND 3 TIER	2136.15	189
85123A	WHITE HANGING HEART T-LIGHT HOLDER	2022.88	776

Interpretation: Identifies strongest revenue products.

Q37. Which products sell the most units?

Perspective: Product Demand and Catalogue Analysis | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT stock_code, description, SUM(quantity) AS total_quantity FROM online_retail_ii WHERE quantity > 0 GROUP BY stock_code,
description ORDER BY total_quantity DESC LIMIT 20;
```

Output sample:

stock_code	description	total_quantity
22693	GROW A FLYTRAP OR SUNFLOWER IN TIN	2424
22086	PAPER CHAIN KIT 50'S CHRISTMAS	1159
85099B	JUMBO BAG RED RETROSPOT	1047
84755	COLOUR GLASS T-LIGHT HOLDER HANGING	848
85123A	WHITE HANGING HEART T-LIGHT HOLDER	776

Interpretation: Identifies high-volume products.

Q38. Which products have the highest average price?

Perspective: Product Demand and Catalogue Analysis | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT stock_code, description, ROUND(AVG(unit_price),2) AS avg_price, COUNT(*) AS lines FROM online_retail_ii WHERE
unit_price > 0 GROUP BY stock_code, description HAVING COUNT(*) >= 2 ORDER BY avg_price DESC LIMIT 20;
```

Output sample:

stock_code	description	avg_price	lines
DOT	DOTCOM POSTAGE	251.03	7
BANK CHARGES	Bank Charges	239.29	3
M	Manual	78.84	4
21627	ELEPHANT CARNIVAL POUFFE	22.21	4
22847	BREAD BIN DINER STYLE IVORY	19.82	5

Interpretation: Compares average product prices.

Q39. Which products appear in the most countries?

Perspective: Product Demand and Catalogue Analysis | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT stock_code, description, COUNT(DISTINCT country) AS country_count FROM online_retail_ii GROUP BY stock_code, description ORDER BY country_count DESC LIMIT 20;
```

Output sample:

stock_code	description	country_count
22551	PLASTERS IN TIN SPACEBOY	5
POST	POSTAGE	5
21700	BIG DOUGHNUT FRIDGE MAGNETS	4
22139	RETROSPOT TEA SET CERAMIC 11 PC	4
22423	REGENCY CAKESTAND 3 TIER	4

Interpretation: Finds products with broad geographic reach.

Q40. Which products have frequent negative quantity rows?

Perspective: Product Demand and Catalogue Analysis | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT stock_code, description, COUNT(*) AS negative_rows FROM online_retail_ii WHERE quantity < 0 GROUP BY stock_code, description ORDER BY negative_rows DESC LIMIT 20;
```

Output sample:

stock_code	description	negative_rows
21231	SWEETHEART CERAMIC TRINKET BOX	2
21232	STRAWBERRY CERAMIC TRINKET BOX	2
21735	TWO DOOR CURIO CABINET	2
22083	PAPER CHAIN KIT RETROSPOT	2
79323GR	GREEN CHERRY LIGHTS	2

Interpretation: Highlights products with frequent returns/corrections.

Q41. Rank products by revenue within each country.

Perspective: Product Demand and Catalogue Analysis | Difficulty: Difficult | SQL concept: Window Function

SQL used:

```
WITH product_country AS (SELECT country, stock_code, description, SUM(line_revenue) AS revenue FROM online_retail_ii WHERE line_revenue > 0 GROUP BY country, stock_code, description) SELECT country, stock_code, description, ROUND(revenue,2) AS revenue, ROW_NUMBER() OVER (PARTITION BY country ORDER BY revenue DESC) AS product_rank_in_country FROM product_country ORDER BY country, product_rank_in_country LIMIT 50;
```

Output sample:

country	stock_code	description	revenue	product_rank_in_country
Australia	23519	EMBROIDERED RIBBON REEL CLAIRE	236.88	1
Australia	20726	LUNCH BAG WOODLAND	145.0	2
Australia	22557	PLASTERS IN TIN VINTAGE PAISLEY	139.2	3

Australia	22580	ADVENT CALENDAR GINGHAM SACK	71.4	4
Australia	22710	WRAP I LOVE LONDON	68.0	5

Interpretation: Ranks products within each country.

Q42. Which products are above average revenue?

Perspective: Product Demand and Catalogue Analysis | Difficulty: Difficult | SQL concept: CTE

SQL used:

```
WITH product_revenue AS (SELECT stock_code, description, SUM(line_revenue) AS revenue FROM online_retail_ii WHERE line_revenue
> 0 GROUP BY stock_code, description), avg_product AS (SELECT AVG(revenue) AS avg_revenue FROM product_revenue) SELECT
stock_code, description, ROUND(revenue,2) AS revenue, ROUND(avg_revenue,2) AS avg_revenue FROM product_revenue, avg_product
WHERE revenue > avg_revenue ORDER BY revenue DESC LIMIT 20;
```

Output sample:

stock_code	description	revenue	avg_revenue
AMAZONFEE	AMAZON FEE	6706.71	50.73
22086	PAPER CHAIN KIT 50'S CHRISTMAS	2988.41	50.73
22693	GROW A FLYTRAP OR SUNFLOWER IN TIN	2286.0	50.73
22423	REGENCY CAKESTAND 3 TIER	2136.15	50.73
85123A	WHITE HANGING HEART T-LIGHT HOLDER	2022.88	50.73

Interpretation: Compares product revenue to average.

Q43. What is the cumulative revenue contribution by product?

Perspective: Product Demand and Catalogue Analysis | Difficulty: Difficult | SQL concept: Window Function

SQL used:

```
WITH product_revenue AS (SELECT stock_code, description, SUM(line_revenue) AS revenue FROM online_retail_ii WHERE line_revenue
> 0 GROUP BY stock_code, description) SELECT stock_code, description, ROUND(revenue,2) AS revenue, ROUND(SUM(revenue) OVER
(ORDER BY revenue DESC ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW),2) AS cumulative_revenue FROM product_revenue ORDER
BY revenue DESC LIMIT 20;
```

Output sample:

stock_code	description	revenue	cumulative_revenue
AMAZONFEE	AMAZON FEE	6706.71	6706.71
22086	PAPER CHAIN KIT 50'S CHRISTMAS	2988.41	9695.12
22693	GROW A FLYTRAP OR SUNFLOWER IN TIN	2286.0	11981.12
22423	REGENCY CAKESTAND 3 TIER	2136.15	14117.27
85123A	WHITE HANGING HEART T-LIGHT HOLDER	2022.88	16140.15

Interpretation: Uses running total to show cumulative product contribution.

Q44. Use product performance view to find leading products.

Perspective: Product Demand and Catalogue Analysis | Difficulty: Difficult | SQL concept: VIEW

SQL used:

```
SELECT stock_code, description, ROUND(total_revenue,2) AS total_revenue, total_quantity FROM v_product_performance ORDER BY
total_revenue DESC LIMIT 20;
```

Output sample:

stock_code	description	total_revenue	total_quantity
AMAZONFEE	AMAZON FEE	6706.71	1

22086	PAPER CHAIN KIT 50'S CHRISTMAS	2988.41	1159
22693	GROW A FLYTRAP OR SUNFLOWER IN TIN	2286.0	2424
22423	REGENCY CAKESTAND 3 TIER	2136.15	189
85123A	WHITE HANGING HEART T-LIGHT HOLDER	2022.88	776

Interpretation: Uses a reusable product performance view.

Q45. Which high-volume products have lower average prices?

Perspective: Product Demand and Catalogue Analysis | Difficulty: Difficult | SQL concept: CTE

SQL used:

```
WITH product_metrics AS (SELECT stock_code, description, SUM(quantity) AS qty, AVG(unit_price) AS avg_price FROM
online_retail_ii WHERE quantity > 0 AND unit_price > 0 GROUP BY stock_code, description) SELECT stock_code, description, qty,
ROUND(avg_price,2) AS avg_price FROM product_metrics WHERE qty > (SELECT AVG(qty) FROM product_metrics) AND avg_price <
(SELECT AVG(avg_price) FROM product_metrics) ORDER BY qty DESC LIMIT 20;
```

Output sample:

stock_code	description	qty	avg_price
22693	GROW A FLYTRAP OR SUNFLOWER IN TIN	2424	1.09
22086	PAPER CHAIN KIT 50'S CHRISTMAS	1159	3.39
85099B	JUMBO BAG RED RETROSPOT	1047	2.75
84755	COLOUR GLASS T-LIGHT HOLDER HANGING	848	0.63
85123A	WHITE HANGING HEART T-LIGHT HOLDER	776	2.94

Interpretation: Finds products with strong volume and lower price.

Q46. Which invoices are cancellations?

Perspective: Returns, Cancellations and Data Quality | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT invoice_no, stock_code, description, quantity, line_revenue FROM online_retail_ii WHERE is_cancelled = 1 LIMIT 20;
```

Output sample:

invoice_no	stock_code	description	quantity	line_revenue
C491012	21231	SWEETHEART CERAMIC TRINKET BOX	-2	-2.5
C491190	21901	KEY FOB , BACK DOOR	-24	-15.600000000000001
C493191	20871	OPULENT VELVET SET/3 CANDLES	-1	-2.1
C493869	85044	RED REINDEER STRING OF 20 LIGHTS	-1	-4.25
C494452	85047	WHITE BEADED GARLAND STRING 20LIGHT	-1	-6.35

Interpretation: Shows cancelled invoice rows.

Q47. Which rows have negative revenue?

Perspective: Returns, Cancellations and Data Quality | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT invoice_no, stock_code, description, line_revenue FROM online_retail_ii WHERE line_revenue < 0 ORDER BY line_revenue
LIMIT 20;
```

Output sample:

invoice_no	stock_code	description	line_revenue
C550456	48185	DOORMAT FAIRY CAKE	-4522.5
C529612	22569	FELTCRAFT CUSHION BUTTERFLY	-508.5
C518904	BANK CHARGES	Bank Charges	-402.69
C525910	M	Manual	-311.99
C506396	BANK CHARGES	Bank Charges	-300.19

Interpretation: Finds returns or credit lines.

Q48. Which rows are missing descriptions?

Perspective: Returns, Cancellations and Data Quality | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT invoice_no, stock_code, description FROM online_retail_ii WHERE description IS NULL OR description = '' LIMIT 20;
```

Output sample:

invoice_no	stock_code	description
495860	16254	
496903	21176	
496955	21772	
498036	16161N	
498693	84653	

Interpretation: Checks product description completeness.

Q49. Which rows are missing country?

Perspective: Returns, Cancellations and Data Quality | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT invoice_no, stock_code, country FROM online_retail_ii WHERE country IS NULL OR country = '' LIMIT 20;
```

Output sample:

invoice_no	stock_code	country

Interpretation: Checks country completeness.

Q50. Which rows have invalid quantity or price?

Perspective: Returns, Cancellations and Data Quality | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT invoice_no, stock_code, quantity, unit_price FROM online_retail_ii WHERE quantity IS NULL OR unit_price IS NULL LIMIT
20;
```

Output sample:

invoice_no	stock_code	quantity	unit_price

Interpretation: Finds records with missing numeric fields.

Q51. Which countries have the most cancellations?

Perspective: Returns, Cancellations and Data Quality | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT country, COUNT(*) AS cancelled_lines FROM online_retail_ii WHERE is_cancelled = 1 GROUP BY country ORDER BY
cancelled_lines DESC LIMIT 20;
```

Output sample:

country	cancelled_lines
United Kingdom	76
France	2
EIRE	2
Spain	1
Italy	1

Interpretation: Counts cancelled rows by country.

Q52. What percentage of rows are cancelled by country?

Perspective: Returns, Cancellations and Data Quality | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT country, COUNT(*) AS total_lines, SUM(is_cancelled) AS cancelled_lines, ROUND(100.0*SUM(is_cancelled)/COUNT(*),2) AS cancellation_pct FROM online_retail_ii GROUP BY country HAVING total_lines >= 20 ORDER BY cancellation_pct DESC LIMIT 20;
```

Output sample:

country	total_lines	cancelled_lines	cancellation_pct
France	72	2	2.78
EIRE	81	2	2.47
United Kingdom	4603	76	1.65
Germany	84	1	1.19
Netherlands	23	0	0.0

Interpretation: Compares cancellation rates.

Q53. Which products have the largest negative revenue?

Perspective: Returns, Cancellations and Data Quality | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT stock_code, description, ROUND(SUM(line_revenue),2) AS net_negative_revenue FROM online_retail_ii WHERE line_revenue < 0 GROUP BY stock_code, description ORDER BY net_negative_revenue ASC LIMIT 20;
```

Output sample:

stock_code	description	net_negative_revenue
48185	DOORMAT FAIRY CAKE	-4522.5
BANK CHARGES	Bank Charges	-702.88
22569	FELTCRAFT CUSHION BUTTERFLY	-508.5
M	Manual	-311.99
21035	SET/2 RED RETROSPOT TEA TOWELS	-275.0

Interpretation: Finds products with largest negative values.

Q54. Which countries have missing customer IDs?

Perspective: Returns, Cancellations and Data Quality | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT country, COUNT(*) AS rows_count, SUM(CASE WHEN customer_id IS NULL THEN 1 ELSE 0 END) AS missing_customer_rows FROM online_retail_ii GROUP BY country ORDER BY missing_customer_rows DESC LIMIT 20;
```

Output sample:

country	rows_count	missing_customer_rows
United Kingdom	4603	1121
EIRE	81	12
Unspecified	5	2
Hong Kong	2	2
France	72	2

Interpretation: Checks customer ID missingness by country.

Q55. Which product descriptions appear against multiple stock codes?

Perspective: Returns, Cancellations and Data Quality | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT description, COUNT(DISTINCT stock_code) AS stock_code_count, COUNT(*) AS lines FROM online_retail_ii WHERE description
<> '' GROUP BY description HAVING stock_code_count > 1 ORDER BY stock_code_count DESC LIMIT 20;
```

Output sample:

description	stock_code_count	lines
found	2	2
TEA TIME OVEN GLOVE	2	7
SET/6 PURPLE BUTTERFLY T-LIGHTS	2	2
SET/3 VANILLA SCENTED CANDLE IN BOX	2	4
SCANDINAVIAN REDS RIBBONS	2	6

Interpretation: Detects catalogue consistency issues.

Q56. Which countries have cancellation rates above average?

Perspective: Returns, Cancellations and Data Quality | Difficulty: Difficult | SQL concept: CTE

SQL used:

```
WITH country_cancel AS (SELECT country, COUNT(*) AS total_lines, SUM(is_cancelled) AS cancelled_lines,
1.0*SUM(is_cancelled)/COUNT(*) AS cancel_rate FROM online_retail_ii GROUP BY country HAVING total_lines >= 20), avg_cancel AS
(SELECT AVG(cancel_rate) AS avg_rate FROM country_cancel) SELECT country, total_lines, cancelled_lines,
ROUND(100*cancel_rate,2) AS cancel_pct, ROUND(100*avg_rate,2) AS avg_cancel_pct FROM country_cancel, avg_cancel WHERE
cancel_rate > avg_rate ORDER BY cancel_rate DESC;
```

Output sample:

country	total_lines	cancelled_lines	cancel_pct	avg_cancel_pct
France	72	2	2.78	1.62
EIRE	81	2	2.47	1.62
United Kingdom	4603	76	1.65	1.62

Interpretation: Compares country cancellation rates to average.

Q57. Rank products by number of cancellation rows.

Perspective: Returns, Cancellations and Data Quality | Difficulty: Difficult | SQL concept: Window Function

SQL used:

```
WITH product_cancel AS (SELECT stock_code, description, COUNT(*) AS cancelled_rows FROM online_retail_ii WHERE is_cancelled =
1 GROUP BY stock_code, description) SELECT stock_code, description, cancelled_rows, RANK() OVER (ORDER BY cancelled_rows DESC)
AS cancellation_rank FROM product_cancel ORDER BY cancellation_rank LIMIT 20;
```

Output sample:

stock_code	description	cancelled_rows	cancellation_rank
21231	SWEETHEART CERAMIC TRINKET BOX	2	1
21232	STRAWBERRY CERAMIC TRINKET BOX	2	1
21735	TWO DOOR CURIO CABINET	2	1
22083	PAPER CHAIN KIT RETROSPOT	2	1
79323GR	GREEN CHERRY LIGHTS	2	1

Interpretation: Ranks products by cancellation frequency.

Q58. Show cancellation trend by month and previous month gap.

Perspective: Returns, Cancellations and Data Quality | Difficulty: Difficult | SQL concept: Window Function

SQL used:

```
WITH monthly AS (SELECT invoice_month, COUNT(*) AS total_lines, SUM(is_cancelled) AS cancelled_lines FROM online_retail_ii
WHERE invoice_month IS NOT NULL GROUP BY invoice_month) SELECT invoice_month, total_lines, cancelled_lines, cancelled_lines -
LAG(cancelled_lines) OVER (ORDER BY invoice_month) AS change_from_previous_month FROM monthly ORDER BY invoice_month LIMIT 20;
```

Output sample:

invoice_month	total_lines	cancelled_lines	change_from_previous_month
2009-12	212	3	None
2010-01	148	3	0
2010-02	138	4	1
2010-03	194	2	-2
2010-04	160	2	0

Interpretation: Uses LAG to compare cancellation counts month to month.

Q59. Use the data quality view to show country completeness.

Perspective: Returns, Cancellations and Data Quality | Difficulty: Difficult | SQL concept: VIEW

SQL used:

```
SELECT country, total_lines, missing_customer_rows, ROUND(missing_customer_pct,2) AS missing_customer_pct FROM
v_country_data_quality ORDER BY missing_customer_pct DESC LIMIT 20;
```

Output sample:

country	total_lines	missing_customer_rows	missing_customer_pct
Hong Kong	2	2	100.0
Unspecified	5	2	40.0
United Arab Emirates	4	1	25.0
United Kingdom	4603	1121	24.35
EIRE	81	12	14.81

Interpretation: Uses a reusable data-quality view.

Q60. Which countries combine high revenue and high cancellation percentage?

Perspective: Returns, Cancellations and Data Quality | Difficulty: Difficult | SQL concept: CTE

SQL used:

```
WITH country_metrics AS (SELECT country, SUM(line_revenue) AS revenue, COUNT(*) AS total_lines, SUM(is_cancelled) AS
cancelled_lines, 1.0*SUM(is_cancelled)/COUNT(*) AS cancel_rate FROM online_retail_ii GROUP BY country HAVING total_lines >=
20) SELECT country, ROUND(revenue,2) AS revenue, total_lines, ROUND(100*cancel_rate,2) AS cancel_pct FROM country_metrics
WHERE revenue > (SELECT AVG(revenue) FROM country_metrics) AND cancel_rate > (SELECT AVG(cancel_rate) FROM country_metrics)
ORDER BY revenue DESC;
```

Output sample:

country	revenue	total_lines	cancel_pct
United Kingdom	83402.03	4603	1.65

Interpretation: Finds markets that are important but potentially risky.

Q61. Which transactions happened in the first invoice month?

Perspective: Time Trends and Operational Planning | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT invoice_no, invoice_date_clean, invoice_month, country, line_revenue FROM online_retail_ii WHERE invoice_month =
(SELECT MIN(invoice_month) FROM online_retail_ii WHERE invoice_month IS NOT NULL) LIMIT 20;
```

Output sample:

invoice_no	invoice_date_clean	invoice_month	country	line_revenue
489434	2009-12-01 07:45:00	2009-12	United Kingdom	83.4
489460	2009-12-01 10:46:00	2009-12	United Kingdom	17.700000000000003
489519	2009-12-01 11:37:00	2009-12	United Kingdom	19.75

489529	2009-12-01 11:51:00	2009-12	United Kingdom	5.0
489536	2009-12-01 12:13:00	2009-12	United Kingdom	3.3

Interpretation: Shows data from the earliest month.

Q62. Which months are present in the dataset?

Perspective: Time Trends and Operational Planning | Difficulty: Easy | SQL concept: SELECT

SQL used:

```
SELECT DISTINCT invoice_month FROM online_retail_ii WHERE invoice_month IS NOT NULL ORDER BY invoice_month LIMIT 50;
```

Output sample:

invoice_month
2009-12
2010-01
2010-02
2010-03
2010-04

Interpretation: Lists months available.

Q63. Which transactions happened during business hours?

Perspective: Time Trends and Operational Planning | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT invoice_no, invoice_hour, description, line_revenue FROM online_retail_ii WHERE invoice_hour BETWEEN 9 AND 17 LIMIT 20;
```

Output sample:

invoice_no	invoice_hour	description	line_revenue
489460	10	FELTCRAFT DOLL MOLLY	17.700000000000003
489519	11	ANT WHITE WIRE HEART SPIRAL	19.75
489529	11	STRAWBERRIES PRINT BOWL	5.0
489536	12	SKULL SHOULDER BAG	3.3
489548	12	PAPER CHAIN KIT 50'S CHRISTMAS	8.850000000000001

Interpretation: Filters transactions by time of day.

Q64. Which transactions happened outside business hours?

Perspective: Time Trends and Operational Planning | Difficulty: Easy | SQL concept: WHERE

SQL used:

```
SELECT invoice_no, invoice_hour, description, line_revenue FROM online_retail_ii WHERE invoice_hour < 9 OR invoice_hour > 17
LIMIT 20;
```

Output sample:

invoice_no	invoice_hour	description	line_revenue
489434	7	15CM CHRISTMAS GLASS BALL 20 LIGHTS	83.4
489666	18	RETRO RED SPOTTY WASHING UP GLOVES	12.600000000000001
490136	19	NURSE'S BAG SOFT TOY	8.95
490143	8	SET OF 20 KIDS COOKIE CUTTERS	23.4
490745	18	IVORY PILLAR CANDLE GOLD FLOCK	6.04

Interpretation: Shows transactions outside common working hours.

Q65. Which weekdays appear in the dataset?

Perspective: Time Trends and Operational Planning | Difficulty: Easy | SQL concept: SELECT

SQL used:

```
SELECT DISTINCT invoice_weekday FROM online_retail_ii WHERE invoice_weekday <> '' ORDER BY invoice_weekday;
```

Output sample:

invoice_weekday
Friday
Monday
Saturday
Sunday
Thursday

Interpretation: Lists transaction weekdays.

Q66. What is monthly revenue?

Perspective: Time Trends and Operational Planning | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT invoice_month, ROUND(SUM(line_revenue),2) AS monthly_revenue, COUNT(*) AS lines FROM online_retail_ii WHERE
invoice_month IS NOT NULL GROUP BY invoice_month ORDER BY invoice_month;
```

Output sample:

invoice_month	monthly_revenue	lines
2009-12	2863.44	212
2010-01	2594.53	148
2010-02	2846.61	138
2010-03	4088.04	194
2010-04	3265.46	160

Interpretation: Aggregates revenue by month.

Q67. Which weekdays have the most transaction lines?

Perspective: Time Trends and Operational Planning | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT invoice_weekday, COUNT(*) AS transaction_lines, ROUND(SUM(line_revenue),2) AS total_revenue FROM online_retail_ii WHERE
invoice_weekday <> '' GROUP BY invoice_weekday ORDER BY transaction_lines DESC;
```

Output sample:

invoice_weekday	transaction_lines	total_revenue
Thursday	952	22922.55
Tuesday	927	16494.39
Monday	879	17017.4
Wednesday	864	19073.16
Friday	721	12949.66

Interpretation: Shows activity by weekday.

Q68. Which hours generate the most revenue?

Perspective: Time Trends and Operational Planning | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT invoice_hour, ROUND(SUM(line_revenue),2) AS revenue, COUNT(*) AS lines FROM online_retail_ii WHERE invoice_hour IS NOT
NULL GROUP BY invoice_hour ORDER BY revenue DESC LIMIT 24;
```

Output sample:

invoice_hour	revenue	lines
10	20319.74	442
11	13475.98	565
12	12390.03	752
15	11097.27	633
14	11059.3	665

Interpretation: Aggregates revenue by hour.

Q69. How does monthly cancellation count change?

Perspective: Time Trends and Operational Planning | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT invoice_month, SUM(is_cancelled) AS cancellations, COUNT(*) AS total_lines FROM online_retail_ii WHERE invoice_month IS NOT NULL GROUP BY invoice_month ORDER BY invoice_month;
```

Output sample:

invoice_month	cancellations	total_lines
2009-12	3	212
2010-01	3	148
2010-02	4	138
2010-03	2	194
2010-04	2	160

Interpretation: Shows monthly cancellations.

Q70. Which country-month combinations generate highest revenue?

Perspective: Time Trends and Operational Planning | Difficulty: Medium | SQL concept: GROUP BY

SQL used:

```
SELECT country, invoice_month, ROUND(SUM(line_revenue),2) AS revenue FROM online_retail_ii WHERE invoice_month IS NOT NULL GROUP BY country, invoice_month ORDER BY revenue DESC LIMIT 20;
```

Output sample:

country	invoice_month	revenue
United Kingdom	2010-11	13348.1
United Kingdom	2011-11	8056.14
United Kingdom	2010-12	6290.6
United Kingdom	2011-09	4825.91
United Kingdom	2010-08	4381.44

Interpretation: Finds high-revenue country-month combinations.

Q71. Show monthly revenue with previous month comparison.

Perspective: Time Trends and Operational Planning | Difficulty: Difficult | SQL concept: Window Function

SQL used:

```
WITH monthly AS (SELECT invoice_month, SUM(line_revenue) AS revenue FROM online_retail_ii WHERE invoice_month IS NOT NULL GROUP BY invoice_month) SELECT invoice_month, ROUND(revenue,2) AS revenue, ROUND(LAG(revenue) OVER (ORDER BY invoice_month),2) AS previous_month_revenue, ROUND(revenue - LAG(revenue) OVER (ORDER BY invoice_month),2) AS monthly_change FROM monthly ORDER BY invoice_month;
```

Output sample:

invoice_month	revenue	previous_month_revenue	monthly_change
2009-12	2863.44	None	None
2010-01	2594.53	2863.44	-268.91
2010-02	2846.61	2594.53	252.08

2010-03	4088.04	2846.61	1241.43
2010-04	3265.46	4088.04	-822.58

Interpretation: Uses LAG to show month-on-month change.

Q72. Calculate running monthly revenue.

Perspective: Time Trends and Operational Planning | Difficulty: Difficult | SQL concept: Window Function

SQL used:

```
WITH monthly AS (SELECT invoice_month, SUM(line_revenue) AS revenue FROM online_retail_ii WHERE invoice_month IS NOT NULL
GROUP BY invoice_month) SELECT invoice_month, ROUND(revenue,2) AS revenue, ROUND(SUM(revenue) OVER (ORDER BY invoice_month
ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW),2) AS running_revenue FROM monthly ORDER BY invoice_month;
```

Output sample:

invoice_month	revenue	running_revenue
2009-12	2863.44	2863.44
2010-01	2594.53	5457.97
2010-02	2846.61	8304.58
2010-03	4088.04	12392.62
2010-04	3265.46	15658.08

Interpretation: Uses a window running total.

Q73. Which months are above average revenue?

Perspective: Time Trends and Operational Planning | Difficulty: Difficult | SQL concept: CTE

SQL used:

```
WITH monthly AS (SELECT invoice_month, SUM(line_revenue) AS revenue FROM online_retail_ii WHERE invoice_month IS NOT NULL
GROUP BY invoice_month), avg_month AS (SELECT AVG(revenue) AS avg_revenue FROM monthly) SELECT invoice_month, ROUND(revenue,2)
AS revenue, ROUND(avg_revenue,2) AS avg_revenue FROM monthly, avg_month WHERE revenue > avg_revenue ORDER BY revenue DESC;
```

Output sample:

invoice_month	revenue	avg_revenue
2010-11	14064.14	3884.64
2011-11	8345.0	3884.64
2010-12	6597.97	3884.64
2011-09	5642.92	3884.64
2011-10	4940.67	3884.64

Interpretation: Uses CTEs to compare each month to the average.

Q74. Use monthly trend view to show revenue trend.

Perspective: Time Trends and Operational Planning | Difficulty: Difficult | SQL concept: VIEW

SQL used:

```
SELECT invoice_month, ROUND(monthly_revenue,2) AS monthly_revenue, total_lines FROM v_monthly_trends ORDER BY invoice_month;
```

Output sample:

invoice_month	monthly_revenue	total_lines
2009-12	2863.44	212
2010-01	2594.53	148
2010-02	2846.61	138
2010-03	4088.04	194
2010-04	3265.46	160

Interpretation: Uses a reusable monthly trend view.

Q75. Rank months by revenue and show cumulative share.

Perspective: Time Trends and Operational Planning | Difficulty: Difficult | SQL concept: Window Function

SQL used:

```
WITH monthly AS (SELECT invoice_month, SUM(line_revenue) AS revenue FROM online_retail_ii WHERE invoice_month IS NOT NULL
GROUP BY invoice_month), total AS (SELECT SUM(revenue) AS total_revenue FROM monthly) SELECT invoice_month, ROUND(revenue,2)
AS revenue, RANK() OVER (ORDER BY revenue DESC) AS revenue_rank, ROUND(100.0*SUM(revenue) OVER (ORDER BY revenue DESC ROWS
BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW)/(SELECT total_revenue FROM total),2) AS cumulative_revenue_pct FROM monthly ORDER
BY revenue_rank;
```

Output sample:

invoice_month	revenue	revenue_rank	cumulative_revenue_pct
2010-11	14064.14	1	14.48
2011-11	8345.0	2	23.07
2010-12	6597.97	3	29.87
2011-09	5642.92	4	35.68
2011-10	4940.67	5	40.77

Interpretation: Ranks months and calculates cumulative revenue share.

5. Closing Reflection

The biggest challenge was turning a large operational dataset into structured analytical questions. The most useful insight is that SQL can quickly highlight where revenue, product demand, customer access, returns, and data-quality risks are concentrated. The project uses clear comments and query-by-query explanations so the logic can be explained in the 1-to-1 review call.